GlucoTwin

A hybrid Digital Twin that warns of dangerous glucose LOWS and HIGHS
2 hours ahead, validated on 100 real patients with Type 2 Diabetes

<Team Name> · <College / Incubator Name>
Happiest Health Digital Twin Challenge 2026

The problem

Reactive diabetes care misses the moments that matter

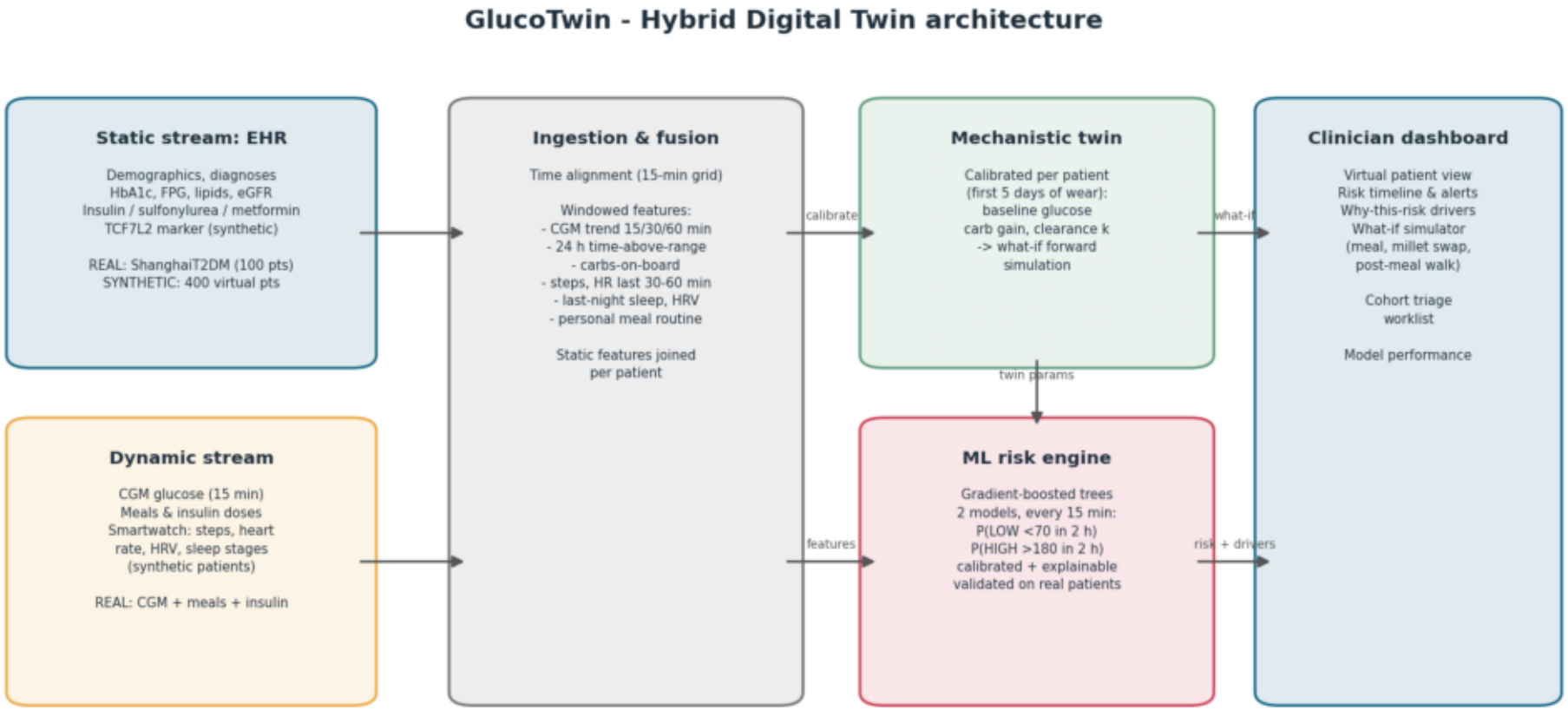
- India has ~101 million people living with diabetes (ICMR-INDIAB, 2023)
- Dangerous LOWS (<70 mg/dL) from insulin & sulfonylureas cause falls, seizures, ER visits
- Repeated HIGHS (>180 mg/dL) after rice-heavy meals drive long-term complications
- Care is reactive: HbA1c every 3 months, occasional glucometer checks
- CGMs & smartwatches produce data every few minutes, but nobody turns it into foresight

Our solution: a hybrid Digital Twin

Predict -> explain -> simulate -> act

- Fuses the static EHR (labs, medicines, genetics) with live CGM, meal, insulin & smartwatch data
- Mechanistic twin: learns each patient's own baseline, carb response and clearance
- Two calibrated ML models re-score LOW and HIGH risk every 15 minutes
- Explainable: shows which data stream drives each alert
- What-if simulator tests lifestyle advice on the twin before it reaches the patient
- Validated on 100 REAL patients (ShanghaiT2DM), not just synthetic data

Architecture



Privacy by design: 100 % synthetic data · no PII · patient-level train/val/test split · DPDP Act / HIPAA sandbox compliant

Stack: Python · NumPy · pandas · scikit-learn · Streamlit · Plotly

Data: real + synthetic, privacy-safe

- REAL: ShanghaiT2DM, 100 people / 109 recordings (Scientific Data 2023, CC BY 4.0)
  - CGM every 15 min, meals with food weights, insulin & oral drugs, labs, complications
- SYNTHETIC: 400 virtual Indian adults x 14 days
  - adds smartwatch streams (steps, heart rate, HRV, sleep) the real data lacks
  - physiology simulator: Indian meals, insulin/sulfonylurea, skipped meals, sleep debt, stress
- No identifiable data; patient-level splits so no patient is in both training and testing

100 REAL patients: LOW glucose (<70) within 2 h

Model	AUROC	Per-patient AUROC	AUPRC	Events	Events warned	Median lead (min)	False alerts/day
Trained on synthetic only	0.683	0.715	0.073	315	14.9%	15	0.13
Trained on real patients	0.859	0.839	0.197	315	81.3%	75	1.25
Trained on synthetic + real	0.849	0.823	0.190	315	87.3%	90	1.66

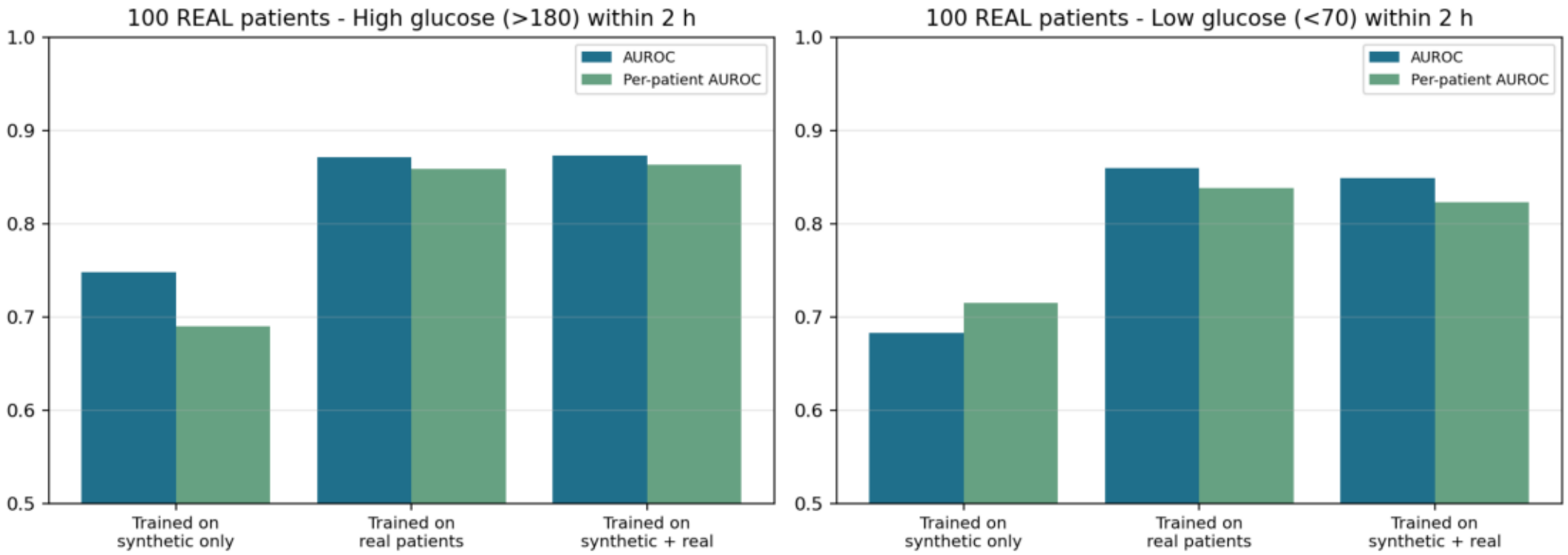
The deployed model warns of 81.3% of real lows, median 75 min ahead (69.2% with 30+ min warning).
5-fold cross-validation grouped by patient: every prediction comes from a model that never saw that patient.

100 REAL patients: HIGH glucose (>180) within 2 h

Model	AUROC	Per-patient AUROC	AUPRC	Events	Events warned	Median lead (min)	False alerts/day
Trained on synthetic only	0.748	0.690	0.339	1482	69.5%	60	1.69
Trained on real patients	0.872	0.859	0.567	1482	91.1%	90	1.85
Trained on synthetic + real	0.873	0.863	0.572	1482	93.2%	90	1.95

The deployed model warns of 93.2% of real spikes, median 90 min ahead.
A model trained on synthetic data alone transfers poorly to real patients: that is why we validate on real data.

Real-patient validation

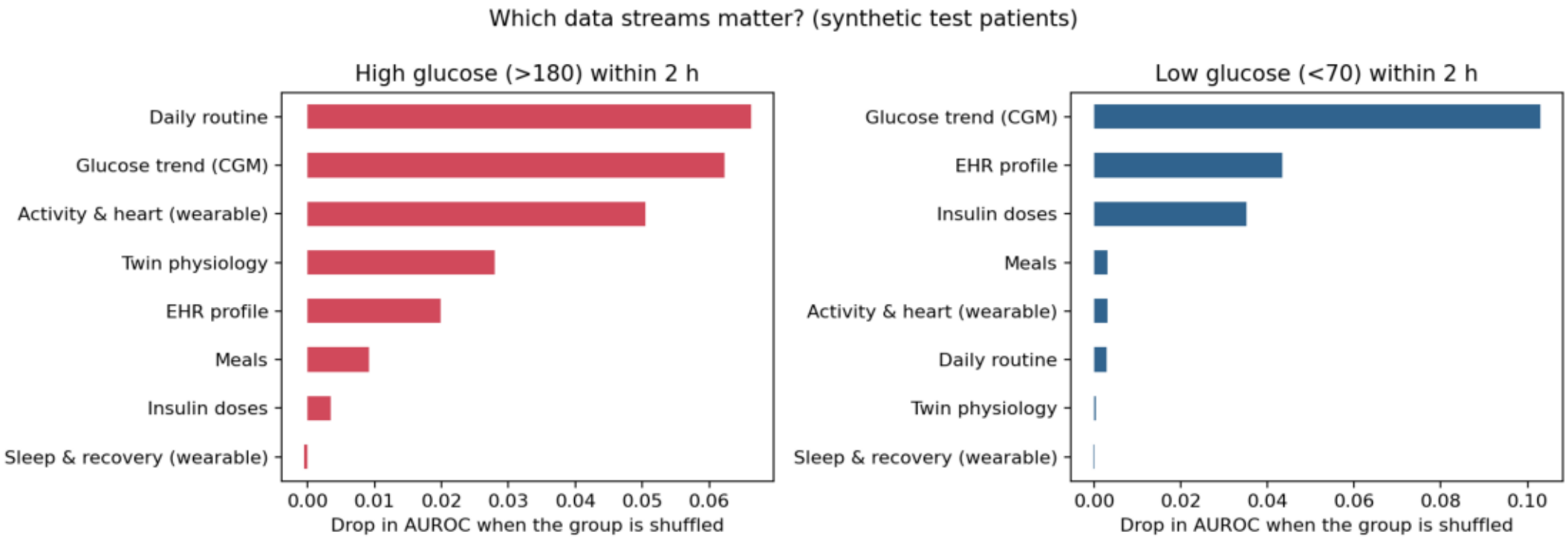


Synthetic cohort: what does each data stream add? (LOW)

Model	AUROC	Per-patient AUROC	AUPRC	Events	Events warned	Median lead (min)	False alerts/day
EHR only (static)	0.952	0.818	0.121	52	50.0%	120	0.17
Wearables + CGM only (dynamic)	0.981	0.962	0.551	52	80.8%	90	0.19
Fused EHR + dynamic	0.985	0.961	0.550	52	86.5%	90	0.2
GlucoTwin (fused + twin)	0.985	0.970	0.539	52	86.5%	75	0.14

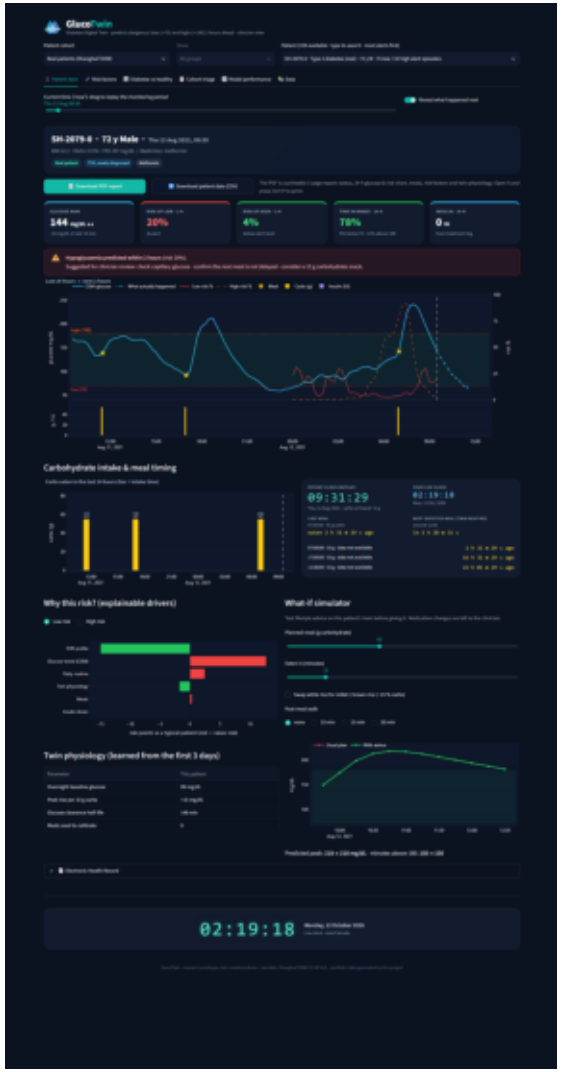
Per-patient AUROC shows whether a model knows WHEN an event comes, not just WHO is at risk.

Which data streams matter?



Drop in AUROC when a feature group is randomly shuffled (synthetic test patients)

Clinician dashboard



Virtual patient · LOW & HIGH risk timeline · explainable drivers · what-if simulator · triage

Healthcare impact

- Doctors: a worklist of who is about to go low or high, not 3-monthly snapshots
- Patients: just-in-time nudges ('eat a 15 g snack now', 'walk 15 min after lunch')
- Health systems: scalable remote monitoring for tier-2/3 India via low-cost CGMs
- Safety-first: medication changes are always left to the clinician

Limitations & roadmap

- Real validation cohort is Chinese and mostly hospitalised: next, validate on Indian CGM cohorts
- Real data has no smartwatch stream: the wearable benefit is shown on synthetic data only
- Carbs estimated from food weights: next, use the Indian Food Composition Tables (IFCT)
- Next: FHIR / ABDM integration, on-device inference, a gestational-diabetes twin
- Clinical pilot with endocrinologists; CDSCO SaMD pathway